Institute of Competitive And Recruitment Exams
(ICARE Academy)

1. **How to Attempt?**

You are developing a feature for a mobile game that tracks player performance. The game records the results of consecutive rounds as a binary array **A** where 1 indicates a win and 0 indicates a loss. The game developers want to analyze how well players are doing by determining the longest streak of consecutive wins. Your task is to find and return an integer value representing the count of the maximum number of consecutive 1s (wins) in the array.

Input Specification:

- **input1**: A binary integer array A representing the game records.
- **input2**: An integer value representing the number of games played.

Output Specification:

Return an integer value representing the count of the maximum number of consecutive 1s (wins) in the array.

Example 1:

- **input1**: {1, 0, 1, 1, 1}
- **input2**: 6
- **Output**: 3

Explanation:

Here, the number of games played is 6 and the game records are {1, 1, 0, 1, 1, 1}. The longest streak of consecutive wins (1s) is 3. Therefore, **3** is returned as the output.

Example 2:

- **input1**: {1, 0, 1, 1, 0, 1}

- **input2**: 6
- **Output**: 2

Explanation:
Here, the number of games played is 6 and the game records are {1, 0, 1, 1, 0, 1}. The longest streak of consecutive wins (1s) is 2. Therefore, **2** is returned as the output.

2. **How to Attempt?**

Riley is working on a new energy calculation project where he needs to track the energy usage of multiple devices. Each device uses energy differently, and Riley needs to compute the alternating energy values for his report. The devices are represented by an integer array **A**, where each element represents the energy consumed by a specific device.

Riley starts by squaring the energy consumption of the first device and adding it to his total. For the second device, he squares its energy consumption and subtracts it from the total. This alternating pattern of addition and subtraction continues for all devices.
Your task is to help Riley find and return an integer value representing the final energy sum after applying this alternating square operation on all devices' energy consumption values.

Input Specification:

- **input1**: An integer value representing the number of devices.
- **input2**: An integer array **A** representing the energy consumed by devices.

Output Specification:

Return an integer value representing the final energy sum after applying the alternating square operation on all devices' energy consumption values.

Example 1:

4
input2 : {2, 3, 4, 1}
Output : 10

Explanation:

Here, there are 4 devices with the energy consumptions as {2, 3, 4, 1}. The sequence of alternating squares is calculated as follows:

- **First device**: 22=42^2 = 422=4 (add)
- **Second device**: 32=93^2 = 932=9 (subtract)
- **Third device**: 42=164^2 = 1642=16 (add)
- **Fourth device**: 12=11^2 = 112=1 (subtract)

The result will be:
4−9+16−1=104 - 9 + 16 - 1 = 104−9+16−1=10

Hence, **10** is returned as the output.

Example 2:

input1: 3
input2: { -5, 6, -2 }

Output : -7

Explanation:

Here, there are 3 devices with energy consumptions as { -5, 6, -2 }. The sequence of alternating squares is calculated as follows:

- **First device**: (−5)2=25(-5)^2 = 25(−5)2=25 (add)
- **Second device**: 62=366^2 = 3662=36 (subtract)
- **Third device**: (−2)2=4(-2)^2 = 4(−2)2=4 (add)

The result will be:
25−36+4=−725 - 36 + 4 = -725−36+4=−7

Hence, **-7** is returned as the output.

3. **How to Attempt?**

There are **N** wrestlers standing in a row with their weights given in an integer array **A**.
You have to find if there is a contiguous segment of exactly **M** wrestlers with the same weight.
Your task is to find such a segment and return an integer value representing the **sum of these M weights**.
If there are multiple such segments, return the **sum of the first one** encountered.
If no such segment exists, return **-1**.

Input Specification:

- **input1**: An integer value **N** representing the number of wrestlers.
- **input2**: An integer array **A** representing the weights of wrestlers.
- **input3**: An integer value **M** representing the number of consecutive wrestlers with the same weight to find.

Output Specification:

Return an integer value representing the **sum of these M weights**.
If there are multiple such segments, return the sum of the **first one** encountered.
If no such segment exists, return **-1**.

Example 1:

- **input1**: 8
- **input2**: {180, 180, 200, 200, 200, 200, 220, 220}
- **input3**: 2

Output: 600

Explanation:
Here, the number of wrestlers is 8 with their weights as
{180, 180, 200, 200, 200, 200, 220, 220}

and **M is 3**, which means we have to find 3 consecutive wrestlers with the same weight.

The contiguous segment of 3 wrestlers with the same weight is {200, 200, 200} and the sum of the segment is **600**. Hence, **600** is returned as the output.

Example 2

- **input1**: 9
- **input2**: {180, 180, 180, 180, 200, 200, 200, 200, 220}
- **input3**: 4
- **Output**: 720

Explanation:
Here, the number of wrestlers is 9 with their weights as
{180, 180, 180, 180, 200, 200, 200, 200, 220}
and **M is 4**, which means we have to find 4 consecutive wrestlers with the same weight.

The contiguous segment of 4 wrestlers with the same weight is {180, 180, 180, 180} and the sum of the segment is **720**.

There exists one more segment {200, 200, 200, 200}, but since it's a **second** segment, it would not be considered.
Therefore, **720** is returned as the output.

4. **How to Attempt?**

You are the organizer of a local book club, and you want to prepare for the next meeting by creating trivia questions based on unique chapter titles from the books your members have read. To do this, you need to analyze a chapter title and count the unique substrings. Your task is to find and return an integer value representing the count of the unique substrings of a specific length **K**.

Input Specification:

- **input1**: A string **S** representing the chapter title
- **input2**: An integer value **K** representing the length of the substrings

Output Specification:

Return an integer value representing the count of the unique substrings of a specific length **K**.

Example 1:

- **input1**: abcabc
- **input2**: 3

Output: 3

Explanation:

Here, the book title is 'abcabc' and **K = 3**. The unique substrings of length 3 that can be formed are:

- "abc"
- "bca"
- "cab"

Since there are 3 unique substrings, **3** is returned as the output.

Example 2:

- **input1**: abcd
- **input2**: 2

Output: 3

Explanation:

Here, the book title is 'abcd' and K = 2. The unique substrings of length 2 that can be formed are:

- "ab"
- "bc"
- "cd"

Since there are 3 unique substrings, 3 is returned as the output.

5. **How to Attempt?:**

Question 1:

You are creating a Feedback Review Page for a new Website. The goal is to style an individual's feedback and display some messages. However, the project is not yet complete, and you need to finish it.

Objectives:

- Create an h2 heading tag with id "heading1" and inner text "Feedback".
- Ensure that the background colour of the button with id submitFeedback is #28a745 and the font color of the button text is white.
- When an individual enters some feedback and clicks on the Submit Feedback button, the message "Thank you for your feedback" should get displayed in the div with id "successMessage".
- Do not change the id or class attributes of any elements, as they are used by the JavaScript code.
- Ensure that the styling is applied correctly according to the provided CSS.

Constraints:

Do not change the id or class attributes of any elements, as they are used by the JavaScript code.

Ensure that the styling is applied correctly according to the provided CSS.

6. **How to Attempt?**

You are building a website, and the client has requested a dynamic dropdown menu for navigation. The menu should be visually styled and should respond to user

interaction. When the user clicks on the "Menu" button, the dropdown list of items should appear. Clicking outside the menu should close the dropdown.

Objective(s):

- Add three list items with text **Home**, **Services**, and **Contact** under the unordered list with id "dropdownList".
- Set the z-index of class dropdown-content to 1.
- Add the javascript functionality to show the dropdown list when clicked on the menu button.

Constraint(s):

- Do not change the "id" of any element as it will cause the code to break.
- Do not remove any existing code, only add the necessary parts.

7. **How to Attempt?**

Write an SQL query to display:

First name, last name, and Account ID of customers who have *ever* made a transaction of strictly greater than 30000. Use alias as "First_Name", "Last_Name", and "Account_ID".

Your output should have 3 columns as given below:

First_Name Last_Name Account_ID

You can view the database schema by clicking the View Schema tab at the bottom of the query window on the right-hand side of the screen.